On Demand
Spain
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Adrián Arnaiz Rodríguez, Ph.D.

Research Scientist

adrianarnaiz.me
Google Scholar
GitHub LinkedIn

Education

PhD, Computer Science, *Cum Laude* — ***Dec 2021 — Sep 2025***
ELLIS Alicante, University of Alicante, Visiting Fellow at Max Planck Institute for Software Systems
- Graduate of the ELLIS PhD Program supervised by **Dr. Nuria Oliver** (ELLIS) and **Dr. Manuel Gómez Rodríguez** (MPI-SWS).
- Trustworthy *high-risk AI decision making* and *social networks* (& GNNs) and it's intersection with the *regulation*.

MSc, AI & Data Science — ***Jan 2020 — Jan 2021***
Universitat Oberta de Catalunya — *GPA:9.7/10*
- Evaluated both Masters and MSc Thesis with Honors.
- **MSc Thesis** supervised by Dr. Baris Kanber from University Collegue of London: *Deep CAE for denoising T1 brain MRI*.

BSc, Computer Engineering — ***Sep 2015 — Jun 2019***
University of Burgos — *GPA:8.9/10*
- Evaluated both Bachelors and BSc Thesis with Honors. Extraordinary award: best student in the promotion.
- **BSc Thesis** at ADMIRABLE research lab: *Voice-biomarkers Parkinson's detection with ML*.

Professional Experience

Research Scientist — ***Oct 2026 — Present***
Apart Research — *Remote*
- We build third-party evaluations of advanced AI, with a focus on assessing risks from AI manipulating people's actions, behaviour, and beliefs.

Postdoctoral Researcher — ***Sep 2025 — Oct 2026***
ELLIS Alicante — *Alicante, Spain*
- Contribute to EU-facing research/proposals on socially-aware GenAI agents, human-AI coevolution, cooperation/competition, orchestration, teaming, opinion dynamics, and emergent collective behavior

Doctoral Researcher — ***Nov 2021 — Sep 2025***
ELLIS Alicante — *Alicante, Spain*
- Trustworthy *high-risk AI decision making* and *social networks* and it's intersection with the *regulation*. Graph ML.
- Funded by Intel Labs (RESUMAIS), ELIAS (European Project [101120237-HORIZON-CL4-2022-HUMAN-02]), Bank Sabadell and GVA.
- Collaborating on the Spanish Charter of Digital Rights under the *Collaboration Agreement C039/23OT* lead by Red.es.

Lecturer on Graph Neural Networks — ***2024***
DelveEducation — *Online*
- Teaching online lessons on Graph Neural Networks (GNNs).

Research Fellow — ***Feb 2024 — Aug 2024***
Max Planck Institute for Software Systems (MPI-SWS) — *Kaiserslautern, Germany*
- Visiting scientist on Human-Centric Machine Learning, advised by Dr. Manuel Gómez Rodríguez.

Research Staff — ***Feb 2020 — Aug 2021***
ADMIRABLE and DATAHES research labs — *University of Burgos, Spain*
- European project [2019-1-ES01-KA204-065615] Self-Regulated Learning: SmartArt.
- Behavioral VR data analysis, Platform *UBUMLaas*, and research on *network science* and *web mining*.

AI Consultant — ***Sep 2019 — Jan 2020***
Ernst & Young — *Madrid, Spain*
- Building ML models for RPA integration: Intelligent Automation

Research Intern — ***Dec 2018 — Jun 2019***
ADMIRABLE research lab — *University of Burgos, Spain*
- Collaboration scholarship. Research in the project *Use of biomarkers extracted from the voice for Parkinson's detection*.

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Honors & Awards

☆ **ELLIS PhD Program Graduate & ELLIS Member** European Laboratory for Learning and Intelligent Systems (ELLIS), 2025 *Academic distinction*. Graduated from the ELLIS PhD Program (Academic Track) after fulfilling all program requirements, including international mobility and first-author publication in a top-tier venue, as established in the ELLIS Letter of Intent.

☆ **Ph.D. Thesis** `Cum Laude` **Distinction** University of Alicante, Sep 2025 *Academic distinction*. Ph.D. thesis evaluated with the highest honors (*cum laude*) by Isabel Valera (Saarland University), Ciro Cattuto (ISI Foundation), and Germán González Serrano (University of Alicante).

☆ **Best Paper and Best Poster Awards** 22nd International Workshop on Mining and Learning with Graphs at ECML-PKDD, Sep 2025 *Competitive award*. Best Paper and poster award at MLG25 workshop at ECML-PKDD 2025. Porto.

☆ **Top Reviewer AISTATS 2025** International Conference on Artificial Intelligence and Statistics, April 2025 *Competitive award*. Selected among the best reviewers.

☆ **London Geometry and Machine Learning Summer School 2024** Imperial College London, July 2024 *Competitive selection*. Selected to take part as a contributor in the project: "GNNs for Relational Databases" with Joshua Robinson.

☆ **Heidelberg Laureate Forum Young Researcher** HLF Foundation, Sep 2023 *Competitive selection*. Selected among 200 young researchers to participate in the 10th HLFF, a highly competitive and prestigious event, where the selected students get the opportunity to interact with Turing Award winners, Abel Prize winners, Fields Medalists, and Nobel Prize winners.

☆ **Top Reviewer AISTATS 2023** International Conference on Artificial Intelligence and Statistics, April 2023 *Competitive award*. Selected in the top 10% best reviewers.

☆ **Top Reviewer ICML 2022** International Conference on Machine Learning, July 2022 *Competitive award*. Selected in the top 10% best reviewers.

☆ **FPU PhD Fellowship (Declined) - FPU20/03687** Spanish Government - Ministry of Universities, Sep 2021 *Competitive PhD Fellowship*. Grants for pre-doctoral contracts in Spanish Universities.

☆ **Extraordinary Award of Bachelors in Computer Engineering** Council of the University of Burgos, Dec 2019 *Competitive award*. Top 1 student in the Degree in Computer Engineering at the University of Burgos.

☆ **Prototype Award** OTRI (Service for Technology Transfer, University of Burgos), Feb 2019 *Competitive award*. Project awarded: Use of bio-markers extracted from the voice for the detection of Parkinson.

☆ **Undergraduate Research Fellowship** University of Burgos, Dec 2018 *Competitive Research Fellowship* from the Ministry of Education of Spain. 6-month internship at the ADMIRABLE research group.

☆ **BSc and MSc Courses with honors** Bachelor's (UBU) and Master's (UOC), 2015-21 8 out of 10 MSc courses evaluated with honors, including Thesis. 10 BSc courses evaluated with honors, including Thesis.

Working Papers

[1] **Adrian Arnaiz-Rodriguez**, Emmanuel Vargas Penagos, Martin Ebers, and Nuria Oliver. "Governing GPAI in Social Platforms: A Purpose-Based Mapping of DSA and AI Act Obligations". In Review.

Publications

[2] **Adrian Arnaiz-Rodriguez**, Miguel Baidal, Erik Derner, Jenn Layton Annable, Mark Ball, Mark Ince, Elvira Perez Vallejos, and Nuria Oliver. "Between Help and Harm: An Evaluation Study of Mental Health Crisis Handling by Large Language Models". In: *JMIR Ment Health* 13 (June 2026), e88435. ISSN: 2368-7959. DOI: 10.2196/88435. URL: https://doi.org/10.2196/88435.

[3] **Adrian Arnaiz-Rodriguez**, Nina Corvelo Benz, Suhas Thejaswi, Nuria Oliver, and Manuel Gomez Rodriguez. "Towards Human-AI Complementarity in Matching Tasks". In: *Machine Learning and Principles and Practice of Knowledge Discovery in Databases*. Accepted as **Oral**. Springer Nature Switzerland, 2026, pp. 36–59. ISBN: 978-3-032-19099-4. DOI: 10.1007/978-3-032-19099-4_3.

[4] **Adrian Arnaiz-Rodriguez** and Federico Errica. "Oversmoothing, "Oversquashing", Heterophily, Long-Range, and more: Demystifying Common Beliefs in Graph Machine Learning". In: *The Fourteenth International Conference on Learning Representations*. 2026. URL: https://openreview.net/forum?id=Q3MisVkuTu.

[5] **Adrián Arnaiz-Rodriguez**, Iana Kazeeva, Kajetan Schweighofer, and Nuria Oliver. "Kann ein Algorithmus fair sein? Technische und rechtliche Perspektiven [Can an Algorithm Be Fair? Technical and Legal Perspectives]". In: *ailex* 3 (Sept. 2026). In German, pp. 123–130. URL: https://rdb.manz.at/document/rdb.tso.LIailex20260305.

[6] **Adrian Arnaiz Rodriguez**. "A Sociotechnical Approach to Trustworthy AI: from Algorithms to Regulation". In: *PhD Thesis, University of Alicante*. 2025. URL: https://ellisalicante.org/publications/arnaiz2025thesis-en/.

[7] **Adrian Arnaiz-Rodriguez**, Georgina Curto Rex, and Nuria Oliver. "Structural Group Unfairness: Measurement and Mitigation by Means of the Effective Resistance". In: *Proceedings of the International AAAI Conference on Web and Social Media*. Vol. 19. 1. June 2025, pp. 83–106. DOI: 10.1609/icwsm.v19i1.35805. URL: https://ojs.aaai.org/index.php/ICWSM/article/view/35805.

[8] Kajetan Schweighofer, **Adrian Arnaiz-Rodriguez**, Sepp Hochreiter, and Nuria Oliver. "The Disparate Benefits of Deep Ensembles". In: *Forty-second International Conference on Machine Learning*. 2025. URL: https://openreview.net/forum?id=tjPxZiqeHB.

[9] **Adrian Arnaiz-Rodriguez** and Ameya Velingker. "*Tutorial on* Graph Learning: Principles, Challenges, and Open Directions". In: *Proceedings of the 41st International Conference on Machine Learning (ICML 2024)*. **Tutorial**. Vienna, Austria, July 2024. URL: https://icml.cc/virtual/2024/tutorial/35233.

[10] **Adrian Arnaiz-Rodriguez** and Julio Losada. "Studying Causality in Algorithmic Decision Making: the Impact of IA in the Business Domain". In: *Revista Internacional y Comparada de Relaciones Laborales y Derecho del Empleo* 12.3 (Dec. 2024). *Written in Spanish*. ISSN: 2282-2313. URL: https://ejcls.adapt.it/index.php/rlde_adapt/issue/view/105.

[11] **Adrian Arnaiz-Rodriguez** and Julio Losada. "The Intersection of Trustworthy AI and Labour Law. A Legal and Technical Study from a Tripartite Taxonomy". In: *Revista General de Derecho del Trabajo y de la Seguridad Social* 69 (Nov. 2024). *Written in Spanish*. ISSN: 1969-9626. URL: https://www.iustel.com/v2/revistas/detalle_revista.asp?id_noticia=427491.

[12] **Adrian Arnaiz-Rodriguez** and Nuria Oliver. "Towards Algorithmic Fairness by means of Instance-level Data Reweighting based on Shapley Values". In: *ICLR 2024 Workshop on Data-centric Machine Learning Research (DMLR)*. Apr. 2024. URL: https://openreview.net/forum?id=ivf1QaxEGQ.

[13] José Luis Garrido-Labrador, José Miguel Ramírez-Sanz, Virginia Ahedo, **Arnaiz-Rodríguez, Adrián**, C. García-Osorio, J. I. Santos, and José Manuel Galán. "Network analysis of co-participation in thesis examination committees in an academic field in Spain". In: *Dirección y Organización*. 79. Asociación para el Desarrollo de la Ingeniería de Organización, 2023, pp. 59–67. DOI: 10.37610/dyo.v0i79.638. URL: https://eventos.uc3m.es/_files/_event/_43190/_editorFiles/file/BookOfAbstracts10.pdf.

[14] **Adrian Arnaiz-Rodriguez**, Francisco Escolano, Nuria Oliver, and Edwin Hancock. "*Tutorial on* Graph Rewiring: From Theory to Applications in Fairness". In: *Proceedings of the Learning on Graphs Conference (LoG 2022)*. **Tutorial**. Virtual Event, Dec. 2022. URL: https://ellisalicante.org/tutorials/GraphRewiring.

[15] **Adrián Arnaiz-Rodríguez**, Ahmed Begga, Francisco Escolano, and Nuria Oliver. "DiffWire: Inductive Graph Rewiring via the Lovász Bound". In: *The First Learning on Graphs Conference*. Dec. 2022. URL: https://openreview.net/forum?id=IXvfIexOmX6f.

[16] Alicia Olivares-Gil, **Arnaiz-Rodríguez, Adrián**, José Miguel Ramírez-Sanz, José Luis Garrido-Labrador, Virginia Ahedo, C. García-Osorio, J. I. Santos, and José Manuel Galán. "Mapping the scientific structure of organization and management of enterprises using complex networks". In: *International Journal of Production Management and Engineering* 10.1 (2022), pp. 65–76. DOI: 10.4995/ijpme.2022.16666. URL: https://polipapers.upv.es/index.php/IJPME/article/view/16666.

[17] **Adrian Arnaiz Rodriguez** and Baris Kanber. "Deep convolutional autoencoders for reconstructing magnetic resonance images of the healthy brain". In: (2021). MSc Thesis. URL: https://hdl.handle.net/10609/127059.

[18] **Adrián Arnaiz-Rodríguez**, José Miguel Ramírez-Sanz, José Luis Garrido-Labrador, and Alicia Olivares-Gil. "Computer Science doctoral research in Spain. A relational perspective". In: *Actas de la XIX Conferencia de la Asociación Española para la Inteligencia Artificial*. Universidad de Málaga. Málaga, 2021, pp. 667–672. ISBN: 978-84-09-30514-8. URL: https://caepia20-21.uma.es/inicio_files/caepia20-21-actas.pdf#page702.

[19] E. Cubo, **Arnaiz-Rodríguez, Adrián**, Álvar Arnaiz-González, J. Díez-Pastor, Meredith Spindler, A. Cardozo, Á. García-Bustillo, Z. Mari, and B. Bloem. "Videoconferencing Software Options for Telemedicine: A Review for Movement Disorder Neurologists". In: *Frontiers in neurology* 12 (2021), p. 1812. DOI: 10.3389/fneur.2021.745917. URL: https://www.frontiersin.org/articles/10.3389/fneur.2021.745917/full.

[20] María-Consuelo Sáiz-Manzanares, Ismael Ramos Pérez, **Adrián Arnaiz-Rodríguez**, Sandra Rodríguez Arribas, Leandro Almeida, and Caroline Martín. "Analysis of the learning process through eye tracking technology and feature selection techniques". In: *Applied Sciences* 11.13 (2021), p. 6157. DOI: 10.3390/APP11136157. URL: https://www.mdpi.com/2076-3417/11/13/6157/htm.

[21] José Luis Garrido-Labrador, José Miguel Ramírez-Sanz, Virginia Ahedo, **Adrián Arnaiz-Rodríguez**, César García-Osorio, José Ignacio Santos, and José Manuel Galán. "Network analysis of co-participation in thesis examination committees in an academic field in Spain". In: *Abstract book of the 14th International Conference on Industrial Engineering and Industrial Management*. 2020. URL: https://eventos.uc3m.es/_files/_event/_43190/_editorFiles/file/BookOfAbstracts10.pdf.

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Leadership & Service to the Scientific Community

— Chair & Organizer roles

[**General Co-Chair**] 5th LoG Conference 2026 ... *Northeastern University, Boston, US, 2026*
[**General Co-Chair**] 4th LoG Conference 2025 ... *Arizona State University, US, 2025*
[**Co-Organizer**] 3rd LoG Conference 2024 ... *Virtual, Dec 2024*
[**Associate Chair**] ICML 2024 Associate Chair ... *Vienna, Jul 2024*
[**Co-Organizer**] Tutorial on *ICML 2024*: "Graph Learning: Principles, Challenges, and Open Directions" ... *Vienna, Jul 2022*
[**Co-Organizer**] Tutorial on *LoG 2022*: "Graph Rewiring: From Theory to Applications in Fairness" ... *Online, Dec 2022*
[**General Co-Chair**] ELLIS Doctoral Symposium 2022 ... *Alicante, Sep 2022*
[**Co-Chair**] ELLIS HCML Reading Group: guest sessions and round tables ... *monthly sessions*
[**Organizer and moderator**] Qualcomm+ELLIS workshop about GNN+HCML ... *Online, Feb 2022*

— Reviewer (PC member)

ICML **2022** 🏆, 2025 (main track and workshops) — AISTATS **2023** 🏆, 2024, **2025** 🏆, 2026 — NeurIPS 2021 – 2026 (main track and workshops) — ICLR 2025, 2026 — FAccT 2023 — ECML-PKDD 2022 — WWW 2022 — ICWSM 2026 — IC2S2 2024, 2025 — MLG 2026 — AIMLSystems 2026

— Journal Reviewer

JMLR — PeerJ — IEEE TNNLS (Graph Special Issue) — Complexity — IEEE TCSS

</> Open Source software

Open source code from all academic projects, open source repositories and contributions at AdrianArnaiz
Main contributor of *Lightly version 1.1.2* in Lightly AI PyTorch-based library for Self-Supervised Learing.

Talks

› Trustworthy AI in Decisions and Networks ... University of Burgos, May 2025
› Round Table on AI for Science ... SENPE Workshop, Pamplona, May 2025
› Algorithmic Fairness and Trustworthy AI ... AI in the government with a gender focus, Alicante, Apr 2025
› AI for High Schoolers: Opportunities and Risks of AI ... Mutxamel, Alicante, Apr 2025
› AI for High Schoolers: Risks of AI Systems Beyond LLMs ... University of Alicante, Mar 2025
› Algorithmic Fairness in High-Stakes AI Systems ... ADItech, Pamplona, Dec 2025
› Trustworthy AI and Regulation ... Ibi The Big Factory, Alicante, Sep 2025
› PhD Thesis Defense ... University of Alicante, Sep 2025
› Towards Human-AI Complementarity in Matching Tasks ... HLDM @ ECMLPKDD 2025, Porto, Sep 2025
› Demystifying Common Beliefs in Graph Machine Learning ... MLG @ ECMLPKDD 2025, Porto, Sep 2025
› Demystifying Common Beliefs of Graph Machine Learning ... GLOW Reading Group, Online, Sep 2025
› Structural Group Unfairness ... ICWSM 2025, Copenhaguen, Jun 2025
› Algorithmic Bias and Discrimination in Labor Law ... Workshop AI and Labour Law, Alicante, May 2025
› Systemic Risk of Decision-Making and Recommender Systems ... Key Opinon Leaders, Alicante, Apr 2025
› Designing Fair and Trustworthy AI Systems ... Joint Research Center of the European Commission, Sevilla, Feb 2025
› IA for Social Good ... Faculty of Philosophy, University of Sevilla, Feb 2025
› IA for Social Good ... Workshop for Tech for Intellectual Disabilities, University of Alicante, Nov 2024
› Graph Neural Networks ... Lecture @ DelveDeepLearning, Online, Sep 2024
› Graph Learning: Principles, Challenges, and Open Directions ... Tutorial @ ICML, Vienna, Jul 2024
› Structural Group Unfairness in Social Networks ... TrustLoG Workshop @ WWW Conference, Singapore, May 2024
› Polarization in Networks and Social Capital ... Guest Lecture @ University of Notre Dame, Indiana, USA, Feb 2024
› Algorithmic Discrimination ... Algorithmic Societies ERC @ CCCB, Barcelona, Nov 2023
› Round Table about Sustainable AI ... Chamber of Commerce, Alicante, Jun 2023
› Technical Challenges of Algorithmic Fairness ... ELLIS-DMMLab (CMU) Workshop, Alicante, Mar 2023
› Algorithmic Fairness for EU Parliamentarians ... European Parliament Committee on Legal Affairs (JURI), Alicante, Feb 2023
› Graph Rewiring: From Theory to Applications in Fairness ... Tutorial @ LoG Conference, Online, Dec 2022
› Fairness in AI: From Classic Approaches to Data Valuation and Social Networks ... SpainAI, Burgos, Dec 2022
› Discrimination in AI and Diversity ... PERTE New Economy of Language, Ministry of Economy,Madrid, Oct 2022
› Fairness in Algorithmic Decision-Making ... University of Burgos, Online, Mar 2022
› ELLIS + Qualcomm AI Research: GNN and HCML ... ELLIS HCML Reading Group, Virtual, Feb 2022
› Fairness in Machine Learning: From Introduction to Current Approaches ... ELLIS Talks, University of Alicante, Jan 2022
› Data Processing in Advanced Learning Technology ... Self-Regulated Learning in SmartArt Learning Conference, Jul 2021

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Teaching & Mentoring

- 2025-01 – **Supervising 2 Erasmus+ Interns**: 1) MSc, Vytautas Magnus University (AI and job market); 2) PhD, SGH Warsaw School of Economics (AI and Recidivism in feedback loops).
- 2024-09 – **Online Course**, Delve Deep Learning (Online) – *Graph Neural Networks*.
- 2024-02 – **Guest Lecture**, University of Notre Dame (IN, USA) – *Polarization in Networks and Social Capital*.
- 2024-07 – **Tutorial**, International Conference on Machine Learning (ICML 2024), Vienna, Austria (with Ameya Velingker (Google Research), +1000 in-person attendees) – *Graph Learning: Principles, Challenges, and Open Directions*. [9]
- 2022-12 – **Tutorial**, Learning on Graphs Conference (LoG 2022), Online (with Ahmed Begga, Francisco Escolano, Nuria Oliver, and Edwin Hancock) – *Graph Rewiring: From Theory to Applications in Fairness*. [14]
- 2022–2025 – **Guest Lectures & Masterclasses** – Delivered sessions on Trustworthy AI, GNNs Algorithmic Fairness, and AI Governance to technical and broad audience.

Institutional Engagements & Collaborations

- **ELLIS Member**, European Laboratory for Learning and Intelligent Systems Network.
- **HLF Alumni**, Heidelberg Laureate Forum Alumni (Young Researcher, 2023).
- **EU projects**: ELLIOT (101214398, role: PostDoc), ELIAS (101120237, PhD Student), SmartArt (2019-1-ES01-KA204-065615, Research Assistant).
- **Spanish Charter of Digital Rights (C039/23OT)**: Algorithmic governance of workplace systems.
- **Industry collaborations**: Intel Labs (RESUMAIS), Banco Sabadell.
- **Academic collaborations**: Google Research, JKU, MPI-SWS, UCL, NEC Labs, University of Notre Dame, University of Nottingham, Derby University, Universidade do Minho, UBU, Cleveland Clinic, University of Pennsylvania, among others.
- **Legal collaborations** with Spanish labour inspectors, policy-makers, and sociologists.

Skills

Languages and libs	Python: ML, DL, GNN and Network Science Libraries (PyTorch, scikit-learn, PyGeometric, NetworkX, Numpy, Scipy, pandas, GurobiPy...), scrapping libraries (selenium, request...), web developing libraries (Flask), algorithmic fairness libraries, LLM/Foundation-model evaluation, Multi-agent experimentation I also developed personal libraries for algorithmic fairness, data valuation and networks. More languages: R, Java, Javascript. - DDBBs: Relational (SQL, MySQL), non-relational (mongo, AWS-boto3...)
AI skills	hands-on AI research experience, critically analyze AI claims, AI evaluation, clear AI communication to large audience, eager to expand public engagement and contribute to AI discussion.
More interests	Mathematical optimization, Mathematical Modeling, Causality, Online Learning
Communication	Spanish (native), English (Fluent)
Soft Skills	Highly Analytical, Detail-Oriented, Strong problem-solving skills, Strategic, Thoughtful Communicator, Interdisciplinary, Resilient, Determined, Ethically Driven, Excellence & Precision, Intellectual Honesty, ambition with pragmatism, Impact & Real-World Relevance

For pictures, and more information, please refer to my webpage. Updated Sep 2026